

Verify Any Ghanaian in Under 3 Seconds.

AI-powered Ghana Card verification.

No agents. No queues. No fraud.

Telecoms

Banking & Fintech

Government

Insurance

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Ghana's Government Has Mandated Digital Identity Verification. Right Now.

The NCA Directive: Ghana's National Communications Authority has mandated SIM card re-registration using the Ghana Card. Every mobile subscriber must be biometrically verified or lose access.

The deadline is not optional. Millions of subscribers. Hundreds of service centres. The current manual process cannot keep up.



40 Million

Mobile subscribers in Ghana who must be re-verified under the NCA directive



Ghana Card

Issued by the National Identification Authority. Already in millions of hands, carries a face photo and unique PIN.



Compliance is Mandatory

Telecoms and service providers are legally obligated to enforce identity verification at SIM registration.

The Current Approach Is Broken at Scale



Manual Agent Verification

A human agent physically inspects the Ghana Card, compares the photo, and types data. Error-prone, slow, and impossible to scale.

Minutes

Avg wait time per customer

0%

Automated fraud detection in manual process



No Face Matching — Fraud Undetectable

Anyone can walk in with someone else's Ghana Card. Without automated face comparison, fraud goes completely undetected.

None

Digital audit trail from manual verifications

40M+

Subscribers to re-verify — beyond manual capacity



No Audit Trail

No tamper-proof record of who was verified, when, or with what confidence. Regulators have no visibility. Telcos have no protection.

Manual verification does not scale to a 40-million-subscriber re-registration exercise. The bottleneck is not the Ghana Card — it is the agent.

INTRODUCING

VeriGH

Instant. Accurate. Tamper-Proof.

Ghana's first cloud-native KYC verification platform.



Instant

Verification result in under 3
seconds



Accurate

AI face matching with 98%+
confidence



Tamper-Proof

Every result written to
immutable audit trail



Scalable

Serverless — handles one or
one million requests

Three Steps. Under Three Seconds.

No app download. No registration. No physical visit. Works on any device with a camera.



STEP 1

Photo the Ghana Card

Customer takes a photo of the front of their Ghana Card using their phone or a device at the service point



STEP 2

Take a Selfie

Customer takes a live selfie. No special equipment needed. A standard smartphone camera is all that is required.



RESULT

Instant PASS / FAIL

AI extracts the name and ID from the card, compares the faces, and returns a verified result in under 3 seconds

Behind the scenes, Amazon Rekognition reads Ghana Card text via OCR, detects faces in both images, and compares them with biometric precision — all within a single serverless API call.

Manual Verification vs. VeriGH

Factor	Manual Agent	VeriGH
Verification time	5–15 minutes	< 3 seconds
Cost per verification	GHS 5–15 (agent + overhead)	< GHS 0.01
Fraud detection	Human eye, unreliable	Biometric AI, 98%+
Concurrent capacity	Limited by agent headcount	Unlimited (serverless)
Audit trail	Paper-based, no digital record	Every record timestamped
Available hours	Service centre hours only	24 / 7 / 365
Location	Must visit a service centre	Any device, anywhere

1,500×

Faster than a manual agent

99%+

Reduction in cost per verification

Zero

Extra infrastructure for 10× traffic spikes

Who Needs This, and Why They Need It Now

Telecommunications Companies

MTN Ghana, Vodafone Ghana, and AirtelTigo are legally required to verify subscribers against the Ghana Card. VeriGH integrates into SIM registration at any service point or kiosk.

Highest urgency

Regulatory deadline

Banks & Financial Institutions

KYC compliance is mandatory under Bank of Ghana regulations. Account opening, loan applications, and digital wallet activation all require verified identity. VeriGH replaces in-branch verification.

High volume

Compliance-driven

Fintechs & Mobile Money Operators

MoMo agents, digital lending platforms, and fintech apps require identity verification at onboarding. A simple API call enables fully remote account activation.

API-first fit

Remote onboarding

Government & Public Services

Voter registration, social welfare disbursements, and e-government portals all require confirmed Ghanaian identity. VeriGH provides the trusted verification layer.

National scale

Policy-aligned

Simple Pricing. Massive Margins at Scale.



Pay-Per-Verification

GHS 0.50

Per verification call. No minimum. No commitment.

Enterprises pay only for what they use.

Infrastructure cost per call is under GHS 0.01,
leaving strong margin.



Enterprise Bundle

Fixed Monthly

Pre-purchased verification volumes at discounted
rates for high-volume clients like telcos and banks.

Predictable revenue, locked-in contracts.



White-Label Licensing

API License

Full API licensing for partners who want to embed
VeriGH inside their own products. Annual license
fee plus per-call royalty.

Unit economics at scale: At 100,000 verifications/month (a single mid-size telco branch network), revenue reaches GHS 50,000/month with infrastructure cost under GHS 1,000.

The Opportunity Is Measurable and Immediate

40M

Subscribers requiring re-verification in
Ghana alone

GHS 20M+

Addressable revenue at GHS 0.50 per
verification

15+

ECOWAS countries with similar SIM
registration mandates

\$1.2B

African digital identity market by 2028
(GSMA)

Ghana is the beachhead. The same platform works across ECOWAS nations using biometric national ID cards. The architecture is already country-agnostic.

Growth Path

1

Ghana: Telcos

MTN, Vodafone, and AirtelTigo SIM re-registration pilots

2

Ghana: Financial Sector

Banks, fintechs, mobile money operators

3

West Africa Expansion

Nigeria, Côte d'Ivoire, and Senegal — NIN, CNPS, and CEDEAO cards

4

Pan-African Platform

Configurable for any biometric national ID document

This Is Not a Simulation. This Is the Real System.

The following results were returned by VeriGH running live on AWS, using a real Ghana Card, a real selfie, and the live Rekognition API.

✅ PASS: Genuine Identity Verified	
Status	PASS ✓
Face Match	98.39%
Name	KWAME ADU MENSAH
ID Number	GHA0087654321
Verification Time	< 3 seconds
Audit Record	Written to DynamoDB ✓

❌ FAIL: Fraud Attempt Detected	
Status	FAIL ✗
Face Match	~11%
Name	Correctly extracted
ID Number	Correctly extracted
Fraud Signal	Selfie ≠ ID photo
Audit Record	Written to DynamoDB ✓

Both PASS and FAIL results are automatically logged to DynamoDB with session ID, timestamp, and confidence score. The audit trail is complete the moment the API call returns.

Built for Regulated Industries. Audit-Ready by Design.



Immutable Audit Trail

Every verification — pass or fail — is written to DynamoDB with session ID, timestamp, confidence score, extracted name, ID number, and outcome. Regulators receive a complete compliance record.



Least-Privilege Security

IAM roles are scoped to the minimum permissions needed. No function has access to resources beyond its specific task. Security built at the infrastructure level, not added on top.



Real-Time Monitoring

CloudWatch alarms trigger the moment any Lambda function throws an error. 14-day log retention. Every API call is traceable end-to-end with full request and response context.



AWS Infrastructure

Runs on Amazon Web Services — trusted by Ghana's largest banks and telecoms. 99.99% SLA. Data residency is configurable to the AWS Africa (Cape Town) region.



Graceful Error Handling

Invalid formats, oversized files, and missing faces all return clean, actionable messages. Raw system errors are never exposed. Every edge case is handled and logged.



Auto-Retry on Throttle

Rekognition API throttling is handled automatically with exponential retry (3 attempts, 2-second delay). No verification fails due to a transient cloud capacity issue.

Serverless Architecture Means No Capacity Ceiling

No Servers to Manage

AWS Lambda scales from 0 to 10,000 concurrent requests without any operator intervention. You pay only for compute you use.

Infrastructure as Code

All AWS resources are defined in Terraform. New environments spin up in minutes. Disaster recovery is a single command.

CI/CD Automation

GitHub Actions deploys every code update automatically. No manual deployments. No stale code in production.

What Powers VeriGH

AI

Amazon Rekognition

Face detection, comparison, and OCR text extraction.

AWS Lambda

AWS Lambda

Serverless Python functions — automatic scaling, pay-per-request.

DynamoDB + S3

DynamoDB + S3

Audit records with millisecond latency and 11-nines durability.

API Gateway

API Gateway

A REST endpoint any client — web, mobile, kiosk — can call.

Roadmap: Closing the Remaining Gaps

NEAR-TERM

Liveness Detection

Add Rekognition Face Liveness to detect whether the selfie is a real person or a printed photo. Closes the last physical spoofing vulnerability.

NIA API Integration

Connect directly to the Ghana National Identification Authority database to cross-validate extracted data against the source of truth.

GROWTH PHASE

ECOWAS Expansion

Extend to Nigerian NIN, Ivorian CNPS, and Senegalese CEDEAO cards. New card types require only updated parsing rules.

Mobile SDK

Native Android and iOS SDKs so fintechs and telcos can embed VeriGH verification directly inside their mobile apps.

PLATFORM VISION

NIA Data Partnership

A formal data-sharing agreement with the National Identification Authority to power verification as a sovereign, government-backed service.

Multi-Sector Expansion

The same Rekognition pipeline supports agricultural disease detection, border control, and healthcare patient verification.

Ghana's SIM Re-Registration Deadline Creates a Narrow Window. We Are Ready.



Pilot Partnership

We are seeking one telco or fintech partner to run a 90-day live pilot with real subscribers, real Ghana Cards, and real compliance data.



API Integration

VeriGH exposes a single REST endpoint. Any existing platform can integrate in hours. We provide SDK support and onboarding.



Regional Distribution

We are open to distribution partnerships for other ECOWAS markets. The platform is configurable for any biometric national ID system.

The demo you just saw is not a prototype. It is a production-ready system running on AWS, deployed with Terraform and continuous delivery, live today. The rest is execution.

THANK YOU

VeriGH

Ghana's Digital Identity Verification Platform

<3s

VERIFICATION

98%+

ACCURACY

\$0.001

PER CALL

40M

MARKET

Questions?

github.com/dkumi12/project-rekognition

rekkye-frontend.s3-website-us-east-1.amazonaws.com

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